

**A Clinical Decision Support System
for Reducing Unnecessary Opioid Prescriptions**

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Abstract

A clinical decision support system (CDSS) is an active knowledge-based system designed to improve healthcare decisions using computer technology. It analyzes patient data in electronic health records, providing prompts to help healthcare providers follow evidence-based guidelines at the point of care. Clinical Decision Support Systems have gained popularity in the past few years in assisting clinicians with various types of complex decision-making. This research paper outlines the process of building and assessing the efficacy of a Clinical Decision Support (CDS) system designed to reduce unnecessary and excessive opioid prescribing. The objective of the CDS system was to guide clinicians in adhering to the latest 2022 Centers for Disease Control and Prevention (CDC) Clinical Practice Guidelines for opioid prescription, comprising 12 recommendations, each with specific criteria. The system was developed using Exsys Corvid software, an interactive expert decision tool. Logic was created based on several if-then-else rules employing various types of variables. The system presented straightforward questions to clinicians intending to prescribe opioids for pain. Based on the responses, it progressed to the next pertinent question until all recommendations were addressed. If a specific criterion was not met, the system provided relevant recommendations. Testing involved 32 case examples, and upon completing each case, the results indicated the system is effective in guiding clinicians to decide whether to initiate opioid prescribing confidently, select the appropriate class of opioid at the correct dose and duration, assess any risk of abuse, and address any potential harms.

Keywords: Clinical Decision Support Systems, outpatient pain management, opioid prescribing, opioid use,

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Objective

The objective is to

1. Describe the development of a CDS system designed to provide clinicians with patient-specific pain management recommendations based on the latest CDC guidelines.
2. Examine the effectiveness of the system in reducing unnecessary opioid prescriptions for pain control in an outpatient setting, advising other treatment options before initiating opioids, recommending appropriate class of opioids, preventing prescribing in higher doses than necessary, mitigating the risk of opioid abuse and overdose, and avoiding the risk of medication errors and adverse events.

Introduction

Opioids are a powerful class of drugs used to reduce pain. These come with both benefits as well as serious risks like drug abuse, overdose, and deaths. Excessive opioid prescribing is a contributing factor to the opioid epidemic or crisis in the USA since the 1990s. In 2021 alone, the country witnessed 750,000 deaths due to opioid overdose, a tenfold increase from 1999.ⁱ Over nearly two decades, the average opioid dispensing rates steadily climbed from 72.4 per 100 individuals in 2006 (approximately 216 million prescriptions) to 81.3 per 100 individuals (approximately 255 million prescriptions) in 2012. Through successful work from entities such as the Centers for Disease Control and Prevention (CDC) and Centers for Medicare and Medicaid (CMS), the rates declined to 43.3 (approximately 142 million opioid prescriptions) in 2020; these numbers are still high enough in some regions for every person to have one equivalent to nine times the average dispensing rates.ⁱⁱ In efforts to combat the Opioid Overdose Epidemic, the CDC took several measures and developed various guidelines, including the latest 2022 CDC Clinical Practice Guideline for Prescribing Opioids for Pain. This research paper is about creating and assessing a Clinical Decision Support (CDS) system to assist clinicians in adhering to CDC guidelines, improving their prescribing practices, and reducing unnecessary opioid prescriptions.

Literature review

- I. Sheryl Spithoff, S. M. (2020). “**Clinical Decision Support Systems for Opioid Prescribing for Chronic Non-Cancer Pain in Primary Care: A Scoping Review**”. *Journal of the American Board of Family Medicine (JABFM)*, 16.

This research looked at information about clinical decision support systems (CDSSs) that help clinicians make decisions about prescribing pain-relieving drugs called opioids. The focus of these systems was on patients with long-term pain not caused by cancer.

The study covered information gathered from various sources from January 1, 2008, to October 11, 2019. Out of 5,068 pieces of information, 14 studies from the United States met the requirements. Some studies were about CDSSs used only in limited places, and others were about

systems connected to programs that monitor the use of prescription drugs. Only three of these computer systems included information based on evidence.

The review points out that there have not been many careful studies on these computer systems for prescribing opioids in regular healthcare for long-term pain. Future studies should use evidence-based information and follow guidelines for studying complex interventions.

II. Robert P. Pierce, B. E. (2022). **“Impact of a Vendor-Developed Opioid Clinical Decision Support Intervention on Adherence to Prescribing Guidelines, Opioid Prescribing, and Rates of Opioid-Related Encounters”**. *Applied Clinical Informatics (ACI)*, 12

This study aimed to evaluate the effects of introducing a clinical decision support system (CDSS) for opioids in a multi-location academic health center. The CDSS was designed to enhance adherence to clinical guidelines and improve prescribing practices, addressing the opioid epidemic in the United States. The research compared data from 30 weeks before and after the CDSS implementation.

The results revealed that the opioid CDSS generated an average of 1,637 alerts per week. Although there were enhancements in the rates of providing naloxone and documenting opioid treatment agreements following CDSS implementation, there was no significant alteration in opioid-prescribing practices or rates of opioid-related encounters. The prescribing outcomes provided by the vendor were in line with those directly derived from the electronic health record (EHR). However, overall, the CDSS did not bring about substantial improvements in these areas.

The study concludes that while the implemented opioid CDSS improved certain aspects of provider adherence to prescribing guidelines, it did not effectively alter opioid-prescribing patterns or influence rates of opioid-related encounters. The authors propose further research to identify the optimal configuration of opioid CDSS for more significant improvements in prescribing practices and patient outcomes.

III. Michelle Nguyen, S. L.-A. (2023). **“Impact of Clinical Decision Support Within the Electronic Medical Record on Opioid Prescribing and Dispensing”**. *The Permanente Journal*, 8

This retrospective study investigated the influence of clinical decision support tools, specifically alerts embedded in the electronic medical record and pharmacy system, on the initial opioid prescribing and dispensing day supply. The research compared a 6-month period before and after the implementation of these alerts in an integrated healthcare system.

The findings revealed that the best practice alert in the electronic medical record led to a significant reduction in the average day supply of opioids from 12.09 to 6.58 days. Likewise, the alert in the pharmacy system resulted in a substantial decrease in the average day supply from 13.46 to 6.96 days.

In summary, the deployment of two clinical decision support tools, per Centers for Disease Control and Prevention recommendations advocating for less than a 7-day supply when initiating opioids, demonstrated a noteworthy decrease in the day supply of opioid prescriptions for opioid-naïve patients at the point of prescribing and dispensing. The findings indicate that such tools can effectively influence prescribers and pharmacists to adopt more cautious opioid prescribing and dispensing practices.

- IV. Barry R. Meisenberg, J. G. (2018, September 28). “**Assessment of Opioid Prescribing Practices Before and After Implementation of a Health System Intervention to Reduce Opioid Overprescribing**”. JAMA Network, 11.

This quality improvement study aimed to measure the effects of multilevel interventions on opioid prescribing within a regional health system. The study compared a 6-month preintervention baseline with a 16-month postintervention period ending in April 2018, analyzing inpatient and outpatient clinical activity. The interventions included prescriber education, enhanced oversight, tools to optimize postoperative discharge prescriptions, reduction of default amounts on standard opioid prescription orders, and patient/public education on opioid risks and alternatives.

The results showed that the total amount of opioid medicine given per patient visit decreased each month. At the end of the study, the amount was 58% lower than before the changes, the amount per opioid prescription was 34% less, and the overall rate of opioid prescriptions was 38% lower.

In summary, the study found that by using various strategies to address patient demand, making clinicians more aware and responsible, and giving access to the right tools, they were able to reduce the overuse of opioids.

Summarizing the Literature

The literature review reveals predominant themes, trends, and patterns indicating a lack of rigorous evaluations for Clinical Decision Support Systems (CDSSs) in the context of opioid prescribing within primary care settings. Consequently, there is a pressing need for future studies to integrate evidence-based components. Results vary, with instances of reduced opioid overprescribing, while in other cases, there is no significant alteration in opioid-prescribing practices or rates of opioid-related encounters.

The Basis for Building the Clinical Decision Support System: *CDC Clinical Practice Guideline for Prescribing Opioids for Pain — United States, 2022*

This guideline provides 12 recommendations for clinicians providing pain care, including those prescribing opioids, for outpatients aged ≥ 18 years. It includes recommendations for managing acute (<1 month), subacute (1–3 months), and chronic (>3 months) pain. The recommendations do not apply to pain related to sickle cell disease, cancer, or patients receiving palliative or end-of-life care. CDC recommends that persons with pain receive appropriate pain treatment, with careful consideration of the benefits and risks of all treatment options in the context of the patient's circumstances. Recommendations are not to be applied as inflexible standards of care across patient populations.ⁱⁱⁱ

The guideline addresses 12 recommendations divided into four areas.

1. Determining Whether or Not to Initiate Opioids for Pain

Recommendation #1 - Nonpharmacologic and Nonopioid Pharmacologic Therapy Consideration

Recommendation #2 - Opioid Therapy Goals Discussion

2. Selecting Opioids and Determining Opioid Dosages

Recommendation #3 - Opioid Immediate Release Form When Starting Opioid Therapy

Recommendations #4 and #5 - Lowest Effective Dose

3. Deciding Duration of Initial Opioid Prescription and Conducting Follow-Up

Recommendation #6 - Prescribe Lowest Effective Dose and Duration

Recommendation #7 - Opioid Therapy Risk Assessment

4. Assessing Risk and Addressing Potential Harms of Opioid Use

Recommendation #8 - Naloxone Consideration

Recommendation #9 - Consider Patient's History of Controlled Substance Prescriptions

Recommendation #10 - Urine Drug Testing

Recommendation #11 - Order Select Variation - Concurrent Use of Opioids and Benzodiazepines

Recommendation #12 - Evidence-based Treatment for Patients with Opioid Use Disorder

Inclusion and Exclusion Criteria for the Clinical Decision Support System

The **Opioid Prescription Control Clinical Decision Support System** was developed with the following criteria:

- 1) It applies when the clinician is prescribing an opioid analgesic with ambulatory misuse potential in the outpatient setting.
- 2) It applies to patients aged 18 years and over.
- 3) It includes recommendations for managing acute (<1 month), subacute (1–3 months), and chronic (>3 months) pain.
- 4) The recommendations do not apply to pain related to sickle cell disease, active cancer, limited life expectancy, or patients receiving palliative or end-of-life care.
- 5) The CDS system was developed with user-friendliness and simplicity of usage, and only the minimum necessary questions to minimize the physician's time using the CDSS. It provides recommendations along with resources and links based on answers.
- 6) Straightforward questions in the form of "yes/no" or multiple choice, numeric values were asked.

Methodology- Development of the Clinical Decision Support System

- 1) This study used Exsys Corvid®, an expert system development tool. It converts the decision-making logic and process of the domain expert into a structured form, which the inference engine can then use to drive interactive sessions that provide advice to end users.
- 2) First two criteria related to provider setting and patients age (as defined above) apply to all patients. Hence, the first two sets of questions/ variables were created to apply to all patients. If these basic two criteria weren't met, the system advised to end the session. If the basic criteria were met, the system proceeded to the next step.
- 3) Two clinical decision trees were developed – one for acute pain and the other for subacute or chronic pain because different sets of recommendations apply to these two categories with some overlap. The first decision tree includes recommendations for acute pain (Recommendations # 1, 3, 4, 5, 6, 8, 9, 11, and 12. The second decision tree includes recommendations for subacute and chronic pain (Recommendations #2, 3, 4, 5, 7, 8, 9, 10, 11, and 12).
- 4) Various variables were defined with appropriate questions or message prompts based on these recommendations with embedded links in some cases.
- 5) I then created logic based on the two decision trees using "heuristic" If/Then rules based on variables to organize and structure the rules in a sequential manner.
- 6) I built a command block that described the procedural flow of system execution to derive the value of all confidence variables and display results in a specific format.
- 7) Once the logic worked, the next step was to design the user interface. I set up Applet to add titles, authors, fonts, colors, positions, and background colors. Corvid Runtime used these settings to control how questions were asked and how results were displayed.
- 8) The interactive session started when a physician wanted to prescribe opioids. The system asked for answers(input) to various questions(variables) from the clinician who wanted to initiate opioid prescriptions. Based on the answers, it then provided patient-specific recommendations.
- 9) When the prescriber didn't follow the CDC guidelines and a particular recommendation was not met, the system alerted the prescriber with applicable recommendations.
- 10) If a recommendation was met, it proceeded to the next question until all the recommendations were met and the prescriber was clear to prescribe the opioid medication.
- 11) Since these recommendations should not be applied as inflexible standards of care across patient populations, there are options to snooze the edit and proceed with opioid prescription even when criteria aren't met.

Questionnaire in the Clinical Decision Support System with answer choices

Initial questionnaire

1. Is the patient Outpatient? (Yes/ No)
2. Is the patient aged 18 or over? (Yes/ No)
3. Is there an evidence of sickle cell disease or limited life expectancy or palliative or end-of-life care or active cancer treatment? (Yes/ No)
4. Which type of pain are you prescribing for? (Acute Pain/ Subacute Pain)

Acute Pain related questionnaire

1. Is the patient with acute pain opioid naive? (i.e. No indication of opioid use in the past 90 days from a prescription, self-report, or a dispensing event) (with “Yes/ No” choices)
2. Which type of Opioid analgesic do you want to prescribe? (with “Immediate Release/Extended-release and long-acting” choices)
3. What is the Morphine MME dose? (Numeric value)
4. How many days are you prescribing for Acute pain? (Numeric value)
5. Has the patient been prescribed Naloxone medications to reduce or reverse opioid overdose? (Yes/ No)
6. Are any of the following factors that increase the risk for opioid overdose present? - 1) Currently receiving benzodiazepine or other central nervous system depressant medications, or 2) High opioid dosages (MME/day ≥ 50), or 3) History of a substance use disorder, or 4) Diagnosed with sleep-disordered breathing condition (Yes/ No)
7. Has Prescription Drug Monitoring Program (PDMP) data been reviewed in the past 90 days, and has the PDMP review procedure been performed in the past 90 days? (Yes/ No)
8. Is there any concurrent medication usage of a benzodiazepine OR other central nervous system depressant medication? (Yes/ No)
9. Any current Opioid Use disorder OR order for Buprenorphine or Methadone medication in the past 90 days? (Yes/ No)

Subacute /Chronic Pain related questionnaire

1. Is the patient with Subacute or Chronic opioid naive? [i.e. 1) No indication of opioid use in the past 90 days from a prescription, self-report, or a dispensing event; 2) Pain management procedure in past 90 days is not present; 3) Pain treatment plan created, edited, or reviewed by the prescribing physician in past 90 days is not present] (Yes/ No)
2. Which type of Opioid analgesic do you want to prescribe? (with “Immediate Release/Extended-release and long-acting” choices)
3. What is the Morphine MME dose? (numeric value)
4. Has Opioid risk-benefit assessment been performed in the past 90 days? (Yes/ No)
5. Has the patient been prescribed Naloxone medications to reduce or reverse opioid overdose? (Yes/ No)
6. Are any of the following factors that increase the risk for opioid overdose present? - 1) Currently receiving benzodiazepine or other central nervous system depressant medications, or 2) High opioid dosages (MME/day ≥ 50), or 3) History of a substance use disorder, or 4) Diagnosed with sleep-disordered breathing condition (Yes/ No)

7. Has Prescription Drug Monitoring Program (PDMP) data been reviewed in the past 90 days, and has the PDMP review procedure been performed in the past 90 days? (Yes/ No)
8. Has urine drug screening (non-opioid or Opioid drug) been performed in the last 12 months? (Yes/ No)
9. Is there a positive result for either of the following in the patient with subacute or chronic pain?- Positive result for a synthetic opioid in the past year and not currently prescribed synthetic opioid OR-Positive result for phencyclidine(PCP) in the past year OR-Positive result for fentanyl in the past year and not currently prescribed fentanyl OR-Positive result for cocaine in the past year OR-Positive result for amphetamine in the past year and not currently prescribed amphetamine OR-Positive result for tetrahydrocannabinol (THC) in the past year and not currently prescribed tetrahydrocannabinol (THC) OR-Positive result for opiate in the past year and not currently prescribed natural opiate OR-Positive result for alcohol in the past year OR-Positive result for methadone in the past year and not currently prescribed methadone(Yes/ No)
10. Is there any concurrent medication usage of a benzodiazepine OR other central nervous system depressant medication? (Yes/ No)
11. Any current Opioid Use disorder OR order for Buprenorphine or Methadone medication in the past 90 days? (Yes/ No)

Alert messages in the Clinical Decision Support System

1. The recommendations only apply to outpatients aged 18 years and older.
2. The recommendations do not apply to pain related to sickle cell disease or cancer or to patients receiving palliative or end-of-life care.
3. Recommendation #1: Recommend the use of nonpharmacologic therapy and nonopioid pharmacologic therapy as alternatives. Nonopioid therapies are at least as effective as opioids for many common types of acute pain
4. Recommendation #2: Maximize Nonopioid Therapies as Appropriate.
5. Recommendation #3: Recommend the use of immediate-release opioids instead of extended-release/long-acting opioids when starting a patient on opioids. Record the reason for opioid therapy or attempt alternative therapy.
6. Recommendations #4 and #5: High risk for opioid overdose - taper now.
7. Recommendation #6: Limit therapy to ≤ 3 days to a maximum of ≤ 7 days and limited to immediate-release opioids.
8. Recommendation #7: Perform assessment of risk for opioid use procedure: Patients on opioid therapy should be evaluated for benefits and harms within 1 to 4 weeks of starting opioid therapy and every 3 months or more subsequently.
9. Recommendation #8: Incorporate into the management plan strategies to mitigate risk; including considering offering naloxone when factors that increase risk for opioid overdose are present. Consider offering naloxone given the risk factor(s).
10. Recommendation #9: PDMP Data Review: Consider Patient's History of Controlled Substance Prescriptions.
11. Recommendation #10: Urine Drug Screening is recommended.

12. Recommendation #11: Avoid prescribing opioid pain medication and benzodiazepine concurrently
13. Recommendation #12: Recommend opioid agonist or partial agonist treatment with methadone maintenance therapy or buprenorphine and/or behavioral therapy
14. All CDC recommendations have been met. Proceed to prescribing the opioid therapy.

Clinical Testing and Evaluation

Testing and validation of application components included:

- Testing the variables for alert messages, links, formatting, and correct values
- Validation of alerts correctly triggered
- Validation of error messages correctly triggered
- Validation of links
- Different screen color for input vs. messages vs. alerts
- Validation of logic working as per the correct sequence and guidelines
- Testing with 32 case scenarios to test the system works according to CDC guidelines

Data and Results

The project used a dataset from 32 case scenarios retrieved from the CDC site to test the system. The case examples included data on the patient's age, status as outpatient or inpatient, name and type of opioid drug prescribed, dosage instructions, dosage duration, last review in days of the Opioid Treatment Care Plan, any concurrent contraindicated medications, evidence of drug abuse, diagnosis of opioid misuse disorder with or without treatment in past 90 days, last urine drug testing. In 19 cases where the clinicians did not follow the guidelines, it correctly flagged the applicable recommendations. In the rest of the 13 cases in which the prescriber followed the guidelines, it correctly cleared the edit and allowed them to proceed with prescribing the opioids. The results indicated that the system works as intended, i.e., correctly flagging all recommendation messages when the prescriber did not follow the guidelines.

Conclusions

The development and testing of the CDS system for opioid prescription control revealed both strengths and limitations.

On the positive side, the system has the potential to significantly reduce opioid prescriptions and enhance compliance with CDC guidelines, thereby improving patient safety and minimizing opioid-related risks such as misuse, overdose, substance use disorder, and deaths. The system can help clinicians make confident decisions, ensuring adherence to relevant regulations and consistency in prescribing practices. Furthermore, there is an opportunity to expand the CDS system to encompass guidelines for prescribing other high-risk drugs.

However, as a standalone system, it has its drawbacks. It cannot access data directly from a patient's medical record. An integrated system connected to the Electronic Health Record (EHR) would be more beneficial for real-time queries and save time for prescribers. Another challenge includes training and motivating providers to use the standalone system alongside the existing medical record system.

A potential area for improvement to this CDS system for Opioid Prescriptions involves adding action choices after each question to provide flexibility for clinicians, such as

- i. Revise the prescription
- ii. Indicate that it is not applicable, log a comment, and snooze for three months
- iii. Indicate that the benefits outweigh the risks and snooze for three months
- iv. Remove - Will remove triggering medication order

This enhancement will require the development of additional logic and commands in the system, which was not in this project's scope.

End Notes

ⁱ Center for Disease Control and Prevention(CDC). “The Drug Overdose Epidemic: Behind the Numbers.” August 8, 2023. Available online at <https://www.cdc.gov/opioids/data/index.html>.

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ⁱⁱⁱ Center for Disease Control and Prevention (CDC). 2022. CDC Clinical Practice Guideline for Prescribing Opioids for Pain — United States, 2022. November 4. Available online at <https://www.cdc.gov/mmwr/volumes/71/rr/rr7103a1.htm?>

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